

EXECUTIVE INVESTIGATION REPORT — CASE MT-001

1. SATELLITE DETECTION & SLICK CHARACTERISTICS

Case Identifier	MT-001
Satellite Sensor	Sentinel-1 C-Band SAR (Interferometric Wide Swath)
Observed Spill Area	14.8 km ² (Radar Backscatter Anomaly)
ML Detection Confidence	94% (Dampened Capillary Wave Signature)
Incident Location	Arabian Sea
Acquisition Time	2026-07-18T06:40:00Z
Discharge Window	2026-07-18T04:10:00Z to 2026-07-18T05:20:00Z

2. DRIFT RECONSTRUCTION & REVERSE ORIGIN MODEL

Reconstructed Origin Center	Lat: 19.3800° N, Lon: 71.1200° E
Origin Confidence Rating	MEDIUM-HIGH
Origin Dispersion Zone	18 km (W) × 11 km (H)
Ocean Surface Currents	1.2 knots (Bearing 115° ESE)
Atmospheric Wind Vector	14.0 knots (Bearing 310° NW)
Dispersion Algorithm	Eulerian-Lagrangian Particle Backward Trajectory Model

3. PRIMARY ATTRIBUTED CANDIDATE DOSSIER

MV OCEAN STAR

MMSI: 636012345 | Type: Oil Tanker | Speed: 11.8 kn | Course: 142°

Proximity to Origin Zone: 19 km

Confidence Classification: HIGH CONFIDENCE ATTRIBUTION

92

COMPATIBILITY / 100

4. MULTI-FACTOR FORENSIC CORRELATION MATRIX

Forensic Dimension	Weighted Score	Evaluation Metric	Status
Spatial Proximity	92 / 100	19 km intercept with origin zone	VERIFIED [PASS]
Temporal Overlap	96 / 100	AIS transmission coincided with slick release window	VERIFIED [PASS]
Trajectory Consistency	89 / 100	Course corridor intersects origin probability centroid	VERIFIED [PASS]
Hydrodynamic Drift	94 / 100	Simulated forward drift footprint matches satellite slick	VERIFIED [PASS]
AIS Transmission Integrity	87 / 100	Continuous broadcasts with zero transponder blackout	VERIFIED [PASS]

5. ALL CORRELATED VESSEL CANDIDATES COMPARISON

Rank	Vessel Name	MMSI	Type	Dist.	Conf.	Spat.	Temp.	Traj.	Drift	Total Score
#1	MV OCEAN STAR	636012345	Oil Tanker	19 km	HIGH	92	96	89	94	92 / 100
#2	MV EASTERN WIND	636045678	Cargo	12 km	MEDIUM	85	60	72	65	73 / 100
#3	MV BLUE HORIZON	636078901	Tanker	25 km	MEDIUM	65	70	55	50	62 / 100
#4	MV CORAL TRADER	636023456	Cargo	22 km	MEDIUM	55	65	40	45	56 / 100
#5	MV SEA FALCON	636034567	Cargo	28 km	LOW	45	50	40	35	47 / 100
#6	MV MERIDIAN	636056789	Tanker	35 km	LOW	30	40	30	20	36 / 100
#7	MV PACIFIC DAWN	636067890	Cargo	40 km	LOW	20	25	20	10	26 / 100

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Chief Maritime Forensic Analyst

Automated Verification Signature